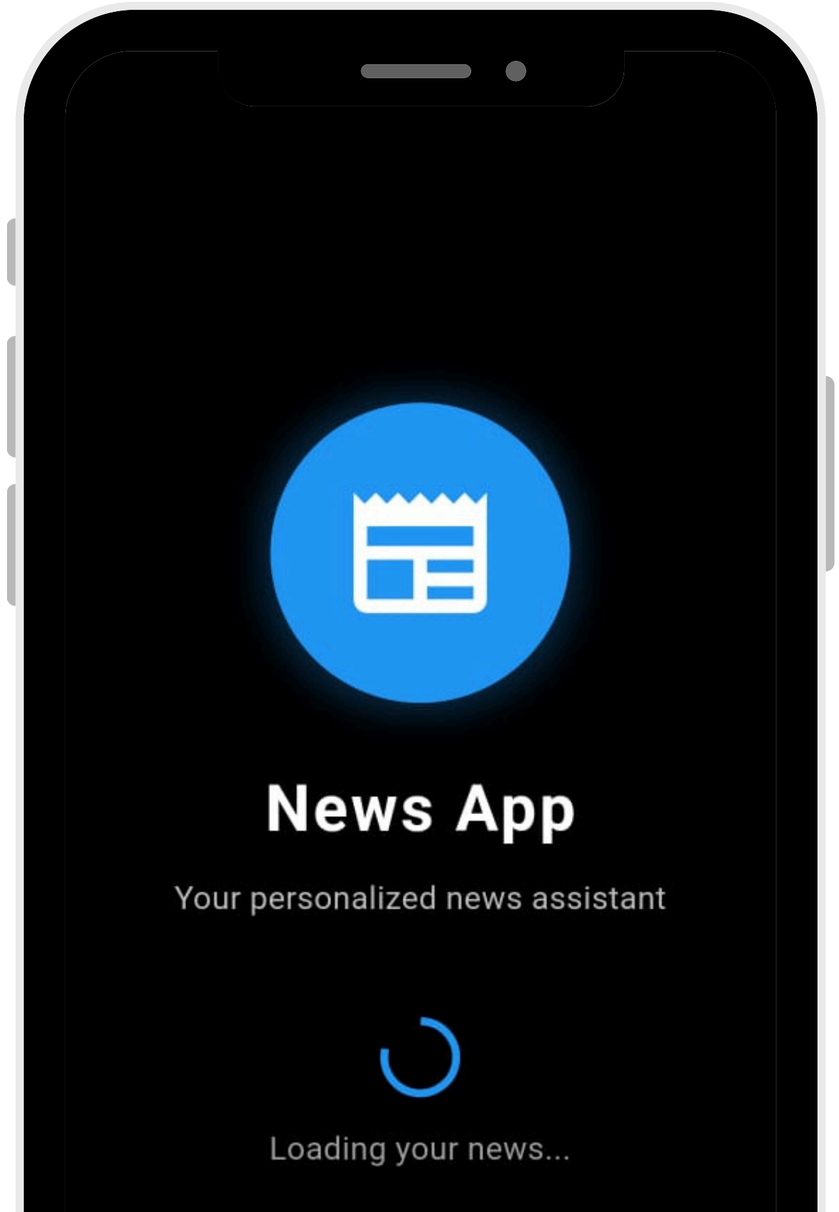
Fake News DETECTION

Explore Now

Play Video



Fake News Detection Using AI

Graduation Project

Student Name:
sarah abutair
shahad khraisheh
lena moqasqas

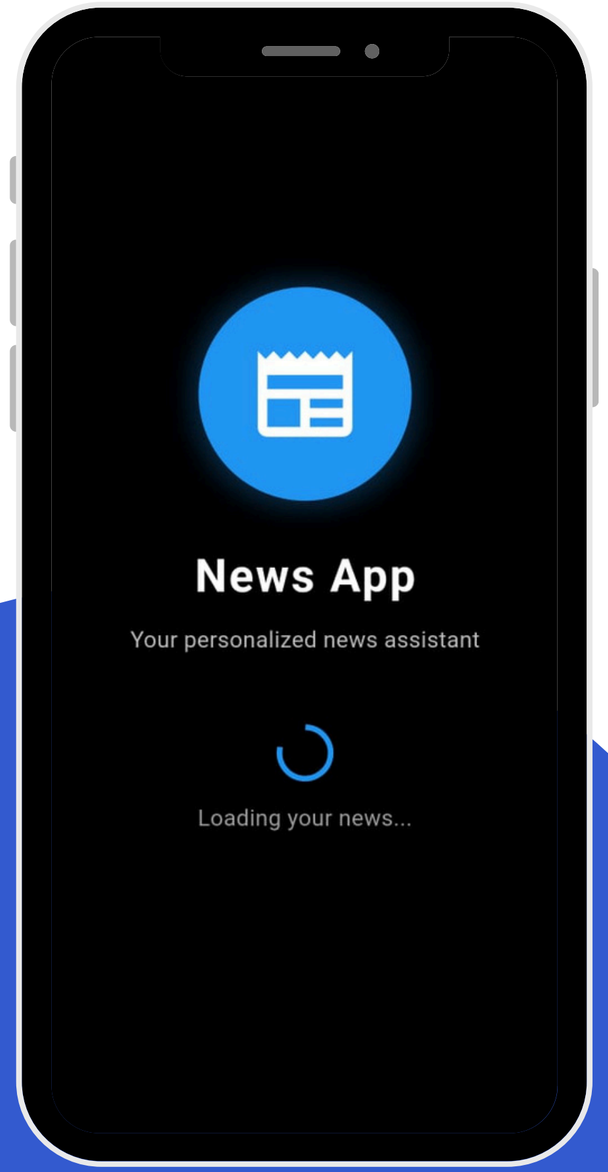
Supervisor:
fatima humor

University: Zarqa University - Faculty of IT

Date: 3/june/2025



Overview



1. Problem Statement
2. Project Aim
3. Objectives
4. Project Limitations
5. Functional & Non-Functional Requirements
6. Context Diagram
7. Data Flow Diagram
8. Graphical User Interface Implementation
9. AI Model Integration
10. Data Training in BERT Model
11. Project Software and Hardware Requirements
12. INTERFACE
13. Features of the App News Assistant (main screen)
14. Testing

Problem Statement

01 Fake news is a major problem in the modern digital world.

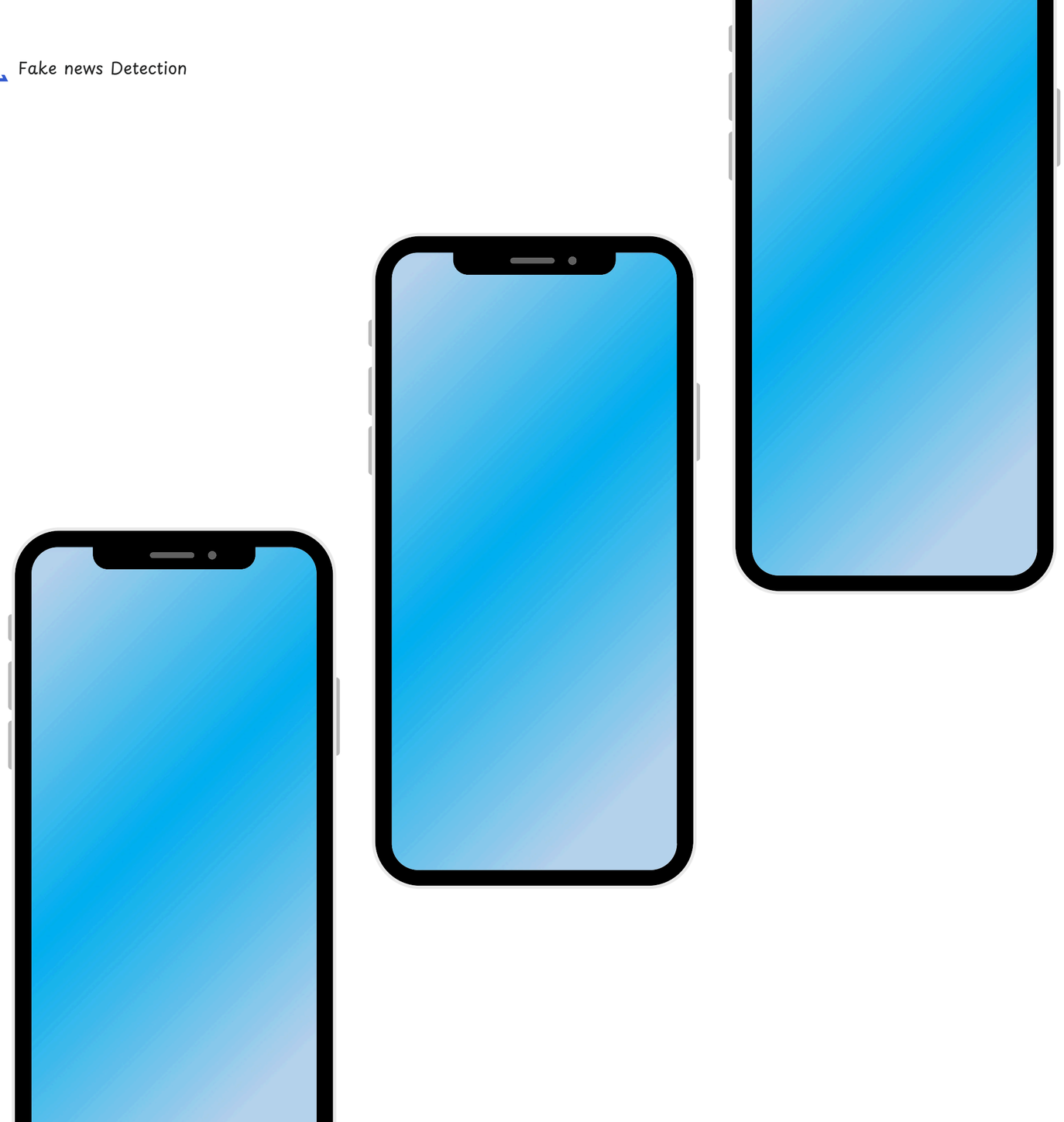
02 Users face difficulty verifying the truth of online articles and headlines.

03 Manual fact-checking is slow and often unreliable due to human bias and limited tools.

04 This project solves the problem by using AI to automatically detect fake news.

04 The system analyzes the text content of articles and user queries to determine credibility.





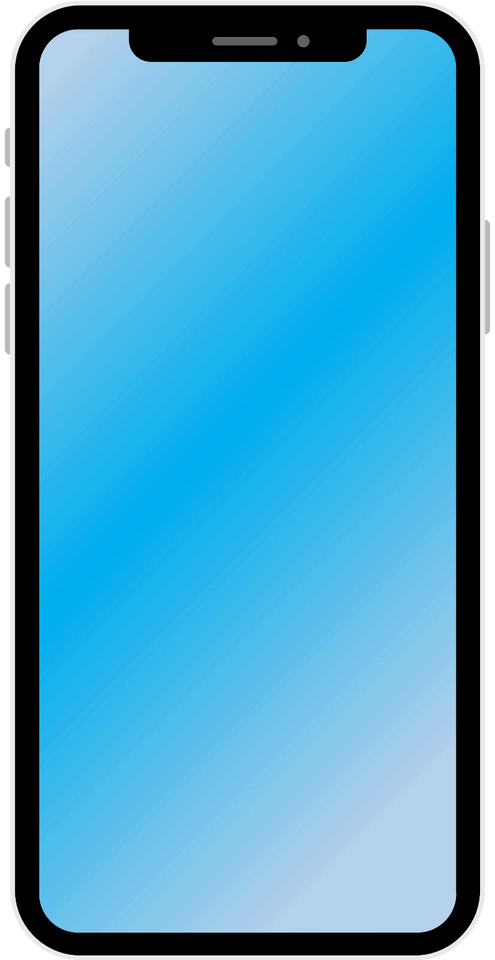
Project Aim

To design and develop a mobile application that uses artificial intelligence to analyze and detect fake news based on content.

Objectives

- Develop a Flutter-based mobile interface for news input and result display
- Use Azure OpenAI to interpret user queries and generate human-like
- Provide real-time feedback on the credibility of the news content

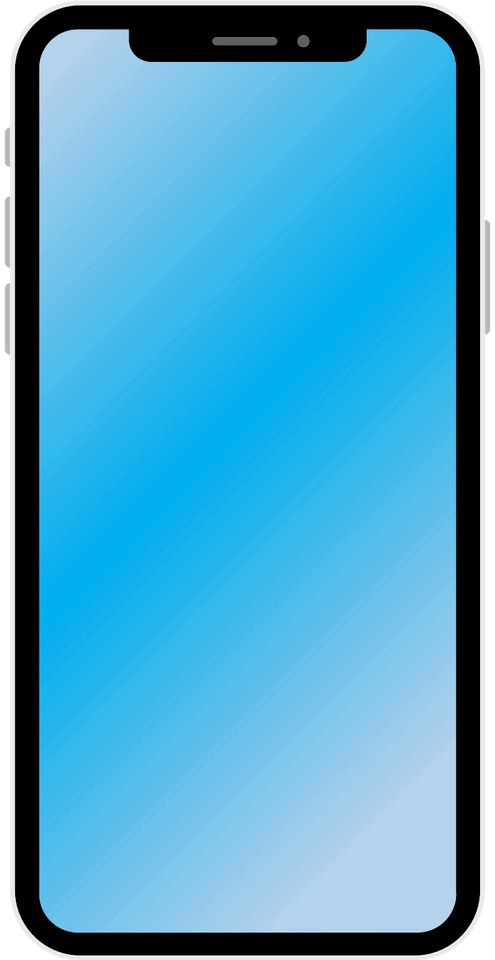
- Integrate a fine-tuned BERT model (Pulk17/Fake-News-Detection) for content classification.
- Deploy the AI model through a RESTful FastAPI backend.
- Store user inputs and results securely using Firebase Realtime Database.

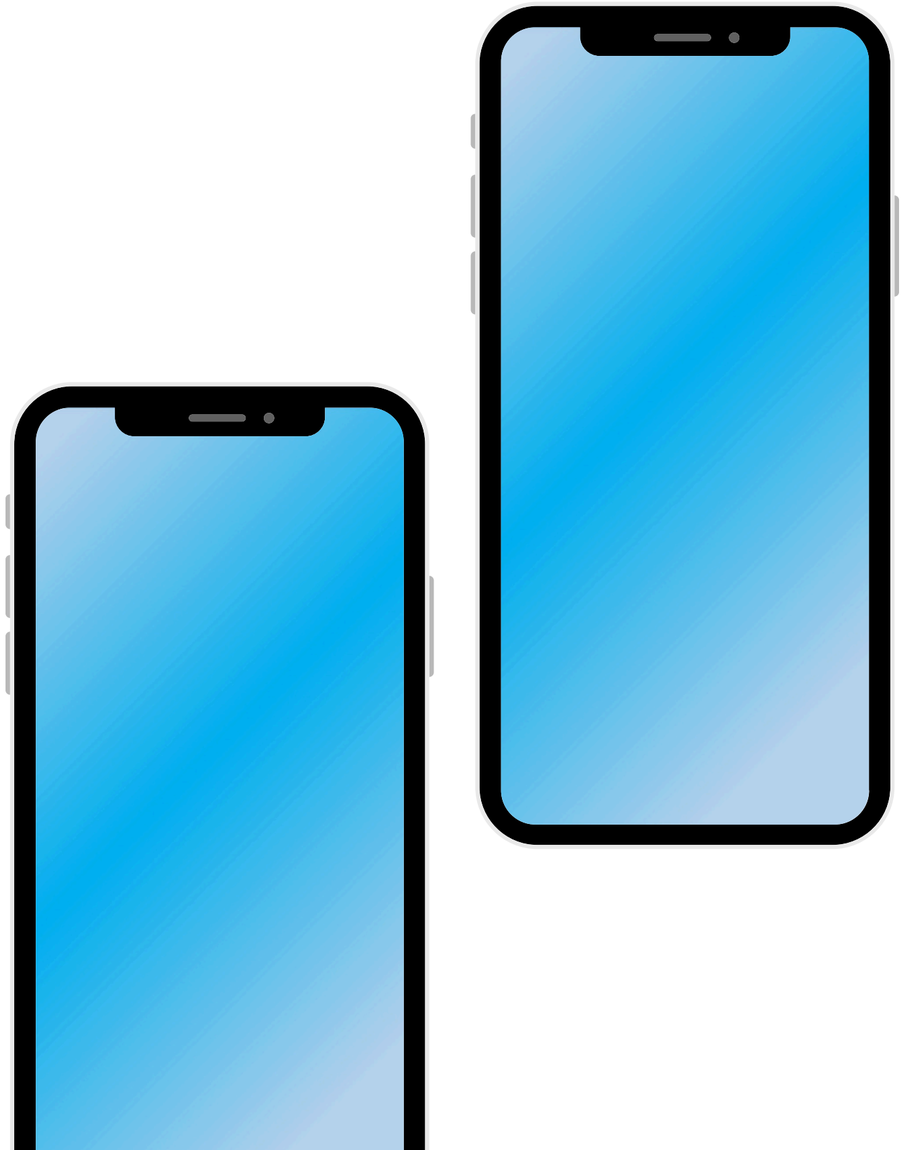


Project Limitations

- The AI model only supports English-language text
- The BERT model is large and cannot be embedded directly in the mobile app

- Requires internet connectivity to function (cloud-based API)
- Does not analyze images or videos—only plain tex





Functional Requirements

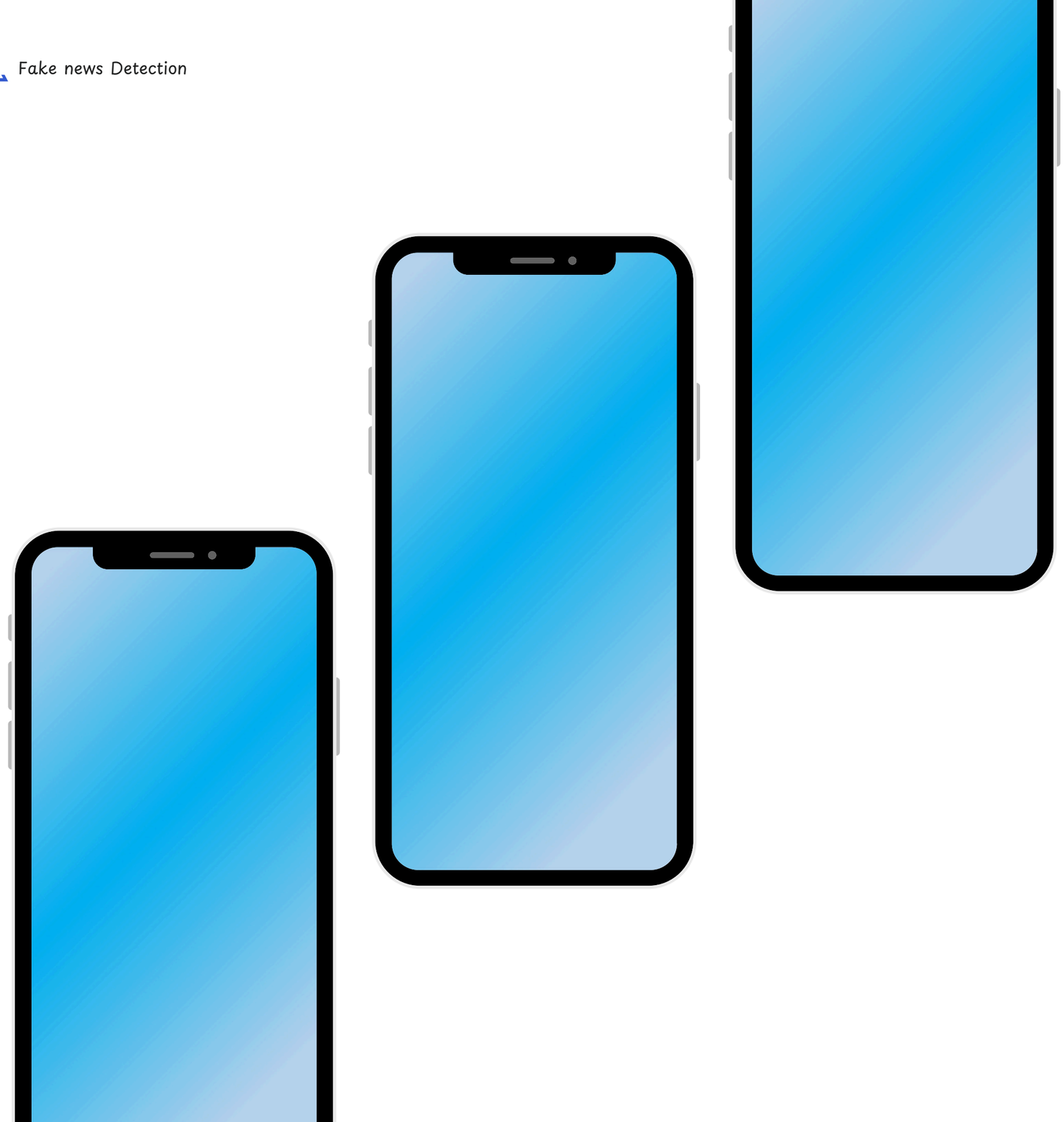
01 User authentication and registration

02 Input text processing (e.g., news headlines or paragraphs)

03 Analysis using a BERT-based fake news classifier

04 Displaying results with confidence scores (Real or Fake)

05 Maintaining conversation history for future reference



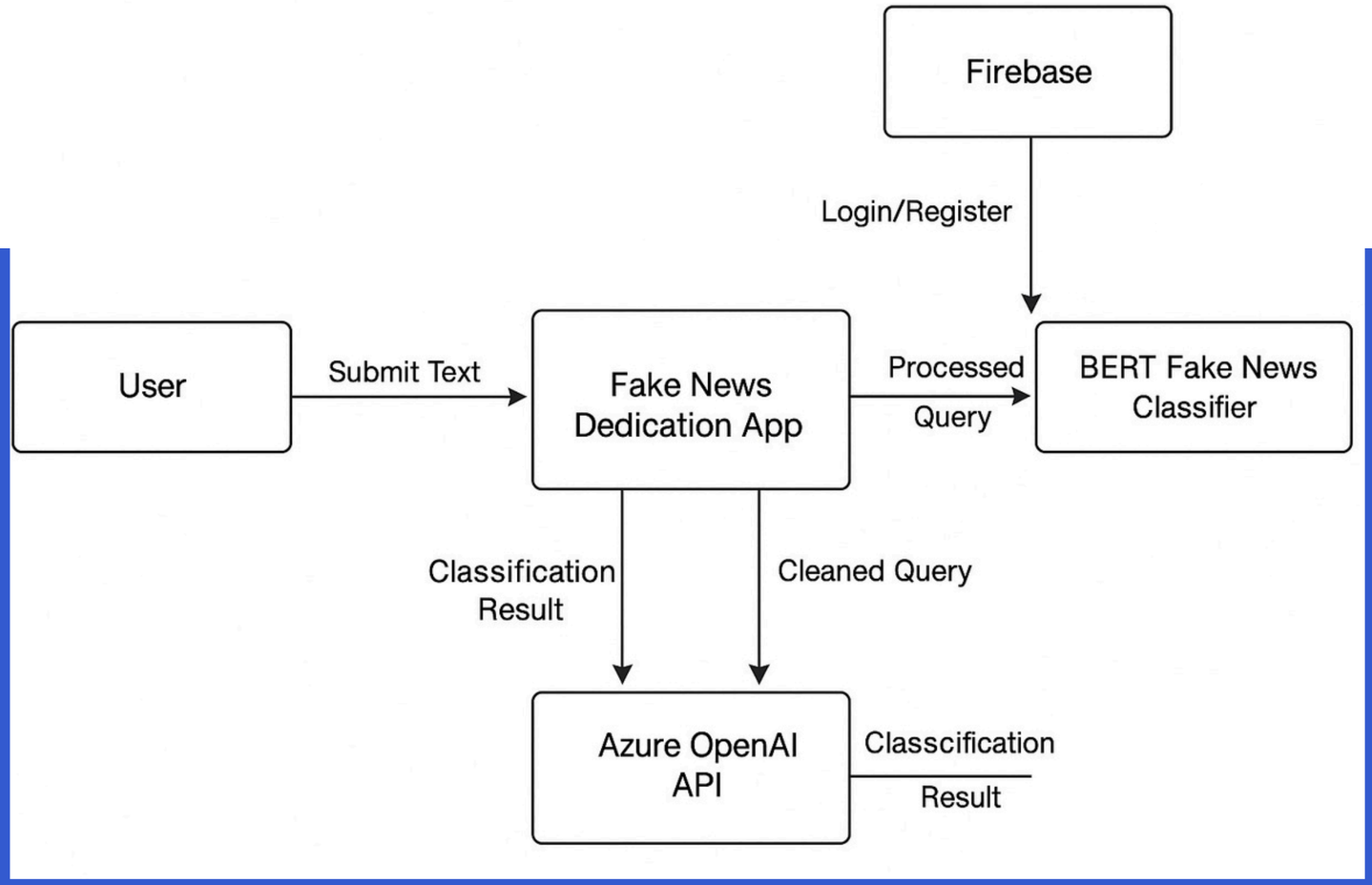
Non-Functional Requirements

01 Performance: The system should return results within 3-5 seconds.

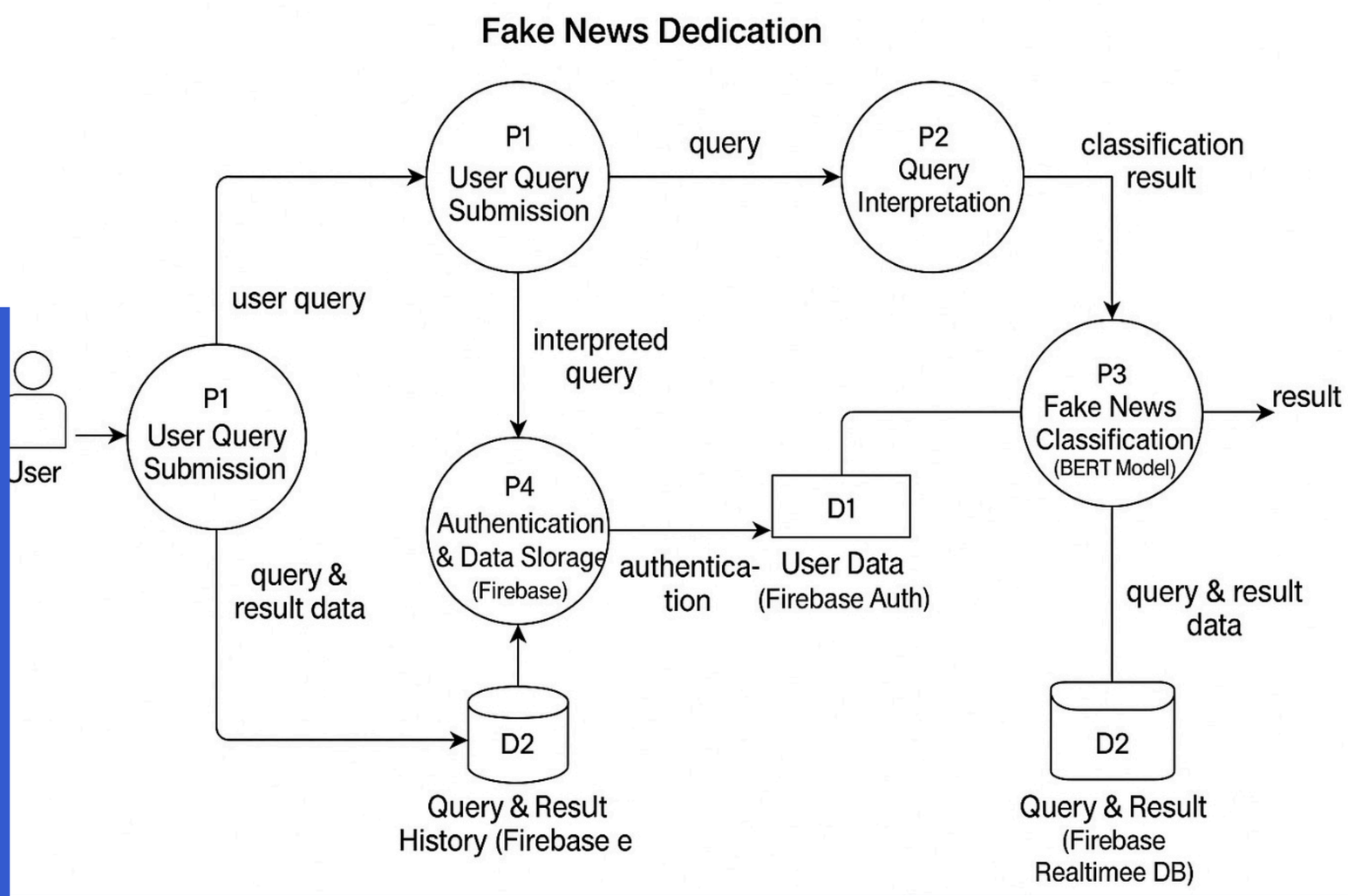
02 Scalability: The backend should handle a moderate number of concurrent users

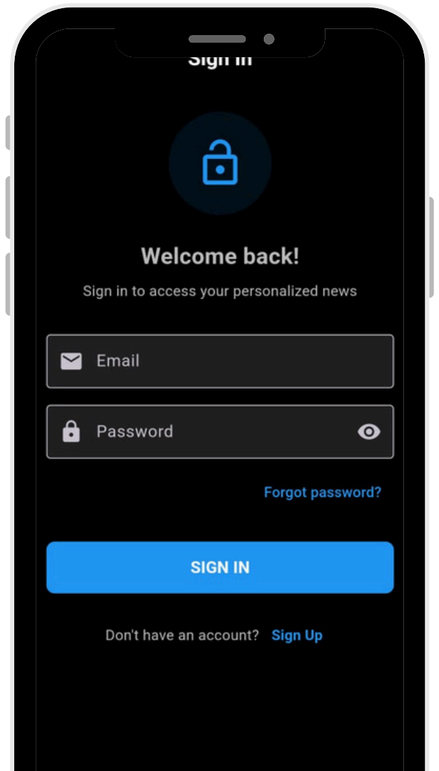
03 Security: Secure user authentication and encrypted communication.

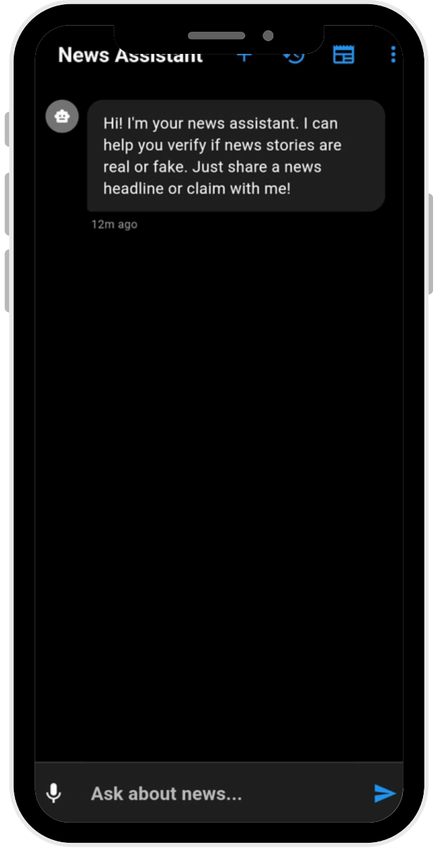
Context Diagram

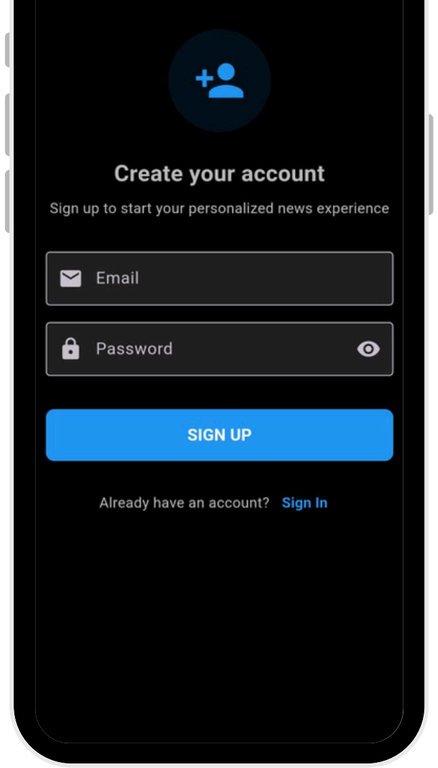


Data Flow Diagram









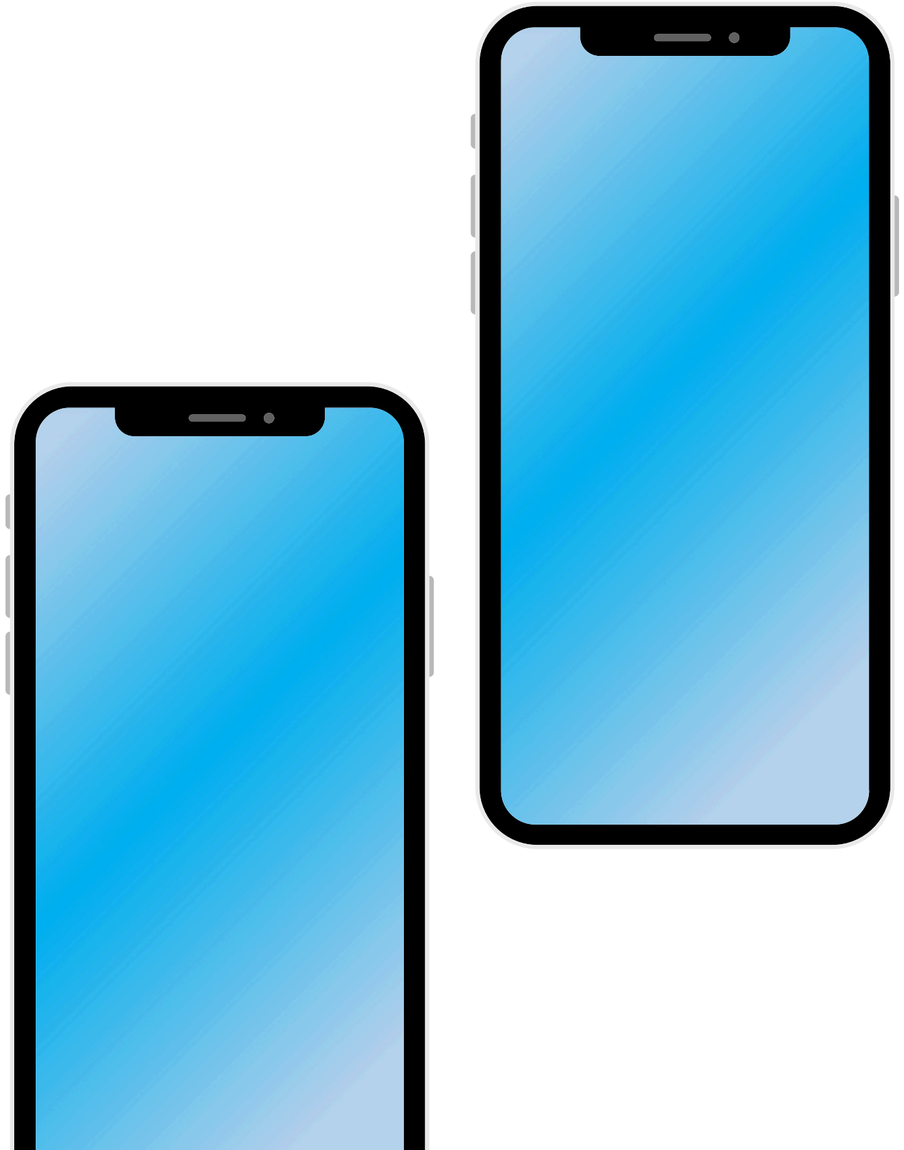
Graphical User Interface Implementation

The frontend is built using Flutter (Dart) to ensure a smooth cross-platform experience on both Android and iOS devices

01 The app interface includes login/signup screens, a main dashboard for submitting news text, and a results screen showing fake news classification with confidence scores

02 The UI follows a clean and minimalistic design approach for usability, with intuitive navigation and clear feedback for user actions

03 the implemented interfaces (high-fidelity prototype) illustrate the user flow from authentication to news checking and history viewing



AI Model Integration

01 The fake news classification model is based on a BERT transformer architecture trained on the Fake-News-Detection dataset (Pulk17) from Hugging Face.

03 The system sends preprocessed input to the model and receives classification predictions with probabilistic scores distinguishing between “Fake” and “Real” news

04 Due to size limitations, the model is deployed as an API hosted on Azure cloud infrastructure

04 Integration with the Azure OpenAI generative model enhances user query understanding, improving classification accuracy

Data Training in BERT Model

```
===== Evaluation Results =====
eval_loss: 0.0160
eval_accuracy: 0.9977
eval_precision: 0.9977
eval_recall: 0.9977
eval_f1: 0.9977
eval_runtime: 28.7408
eval_samples_per_second: 104.3810
eval_steps_per_second: 6.5410
epoch: 3.0000
```

[10125/10125 38:31, Epoch 3/3]

Step	Training Loss
500	0.111100
1000	0.066000
1500	0.039600
2000	0.034900
2500	0.040000
3000	0.049200
3500	0.031600
4000	0.016100
4500	0.009900
5000	0.017200
5500	0.036900
6000	0.014300
6500	0.020100
7000	0.019700
7500	0.005000
8000	0.010200
8500	0.003600
9000	0.012600
9500	0.003200
10000	0.002600

```
===== PREDICTIONS =====

Text 1: BREAKING: Drinking coffee cures cancer, says Nobel-winning scientist..
Prediction: Real
Confidence: 1.0000 (100.0%)

Text 2: The World Health Organization (WHO) today issued new guidelines limiti
Prediction: Real
Confidence: 0.9731 (97.3%)

Text 3: NASA confirms Earth is flat after 50 years of cover-up....
Prediction: Real
Confidence: 0.9869 (98.7%)

Text 4: Local teacher wins national award for innovative STEM curriculum....
Prediction: Real
Confidence: 0.9921 (99.2%)

Text 5: Study: Eating chocolate every day leads to Alzheimer's....
Prediction: Fake
Confidence: 0.9995 (99.9%)

Text 6: Reuters: Global oil prices dip 2% amid supply concerns...
Prediction: Fake
Confidence: 0.9971 (99.7%)
...

Text 12: TechCrunch: Startup X raises $20 million in Series A to build AI-powe
Prediction: Real
Confidence: 1.0000 (100.0%)
```

Project Software and Hardware Requirements

- Flutter (Dart): For building the cross-platform mobile application interface
- Python with FastAPI: For backend development and API creation.
- Azure OpenAI APIs: For handling user query interpretation through a generative language model.
- Hugging Face Transformers: To integrate the BERT-based fake news classification model

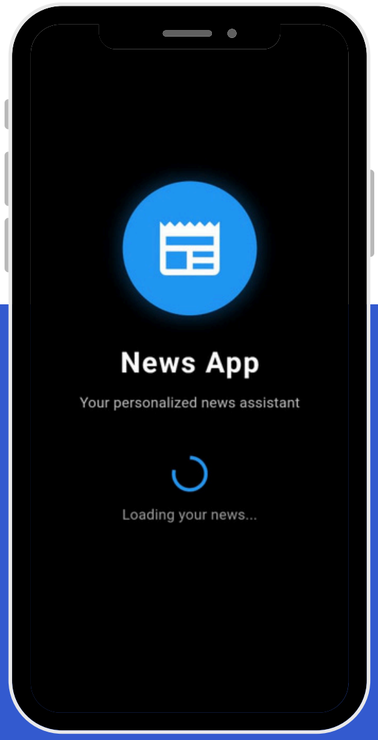
Project Software and Hardware Requirements

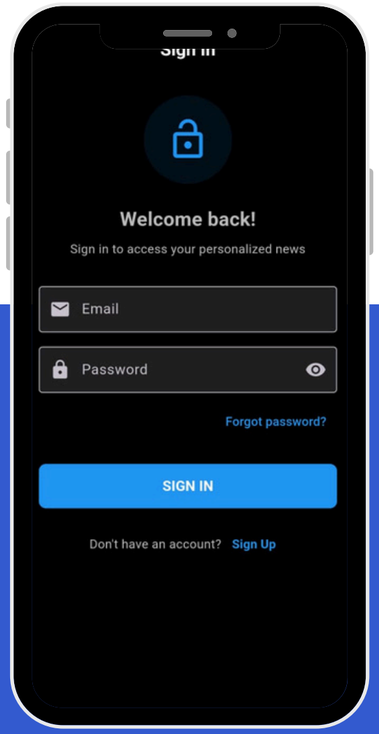
- **Firebase SDKs: For user authentication and real-time database functionalities.**
- **Android Studio & Visual Studio Code: Primary development environments for frontend and backend respectively**
- **Postman & Swagger (OpenAPI): For testing and documenting RESTful API endpoints**
- **Fake-News-Detection-dataset (Pulk17): A labeled dataset from Hugging Face used for training and evaluating the BERT model**

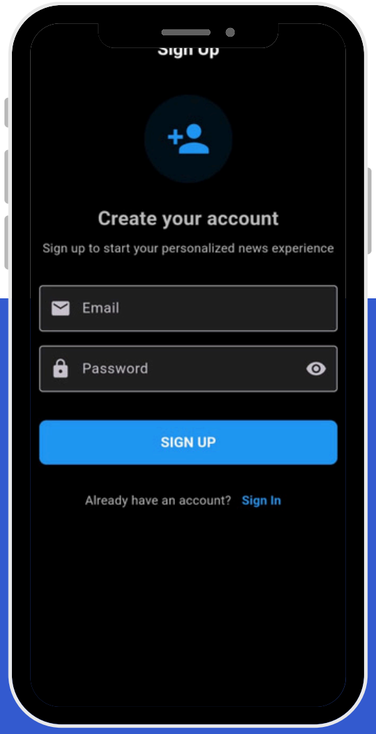
Project Software and Hardware Requirements

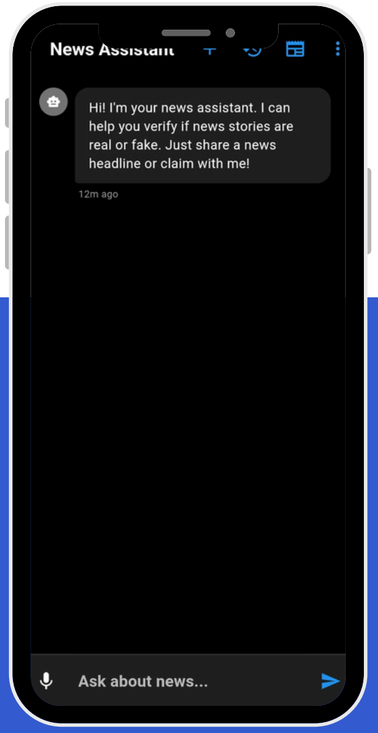
- A smartphone (Android/iOS) for mobile application testing
- A personal computer with internet access for development and deployment.
- Cloud infrastructure (Azure) to host and run the AI model via API

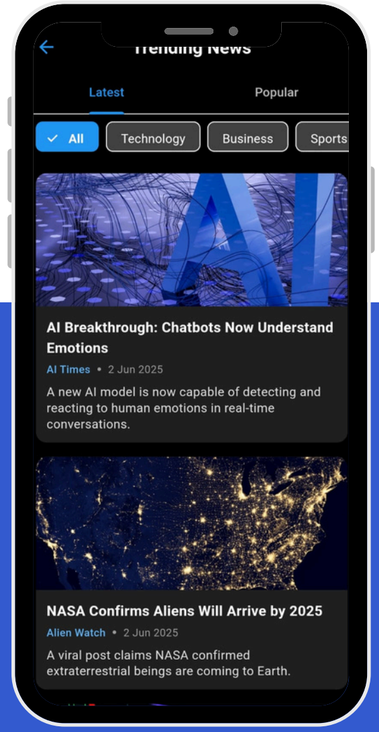
INTERFACE

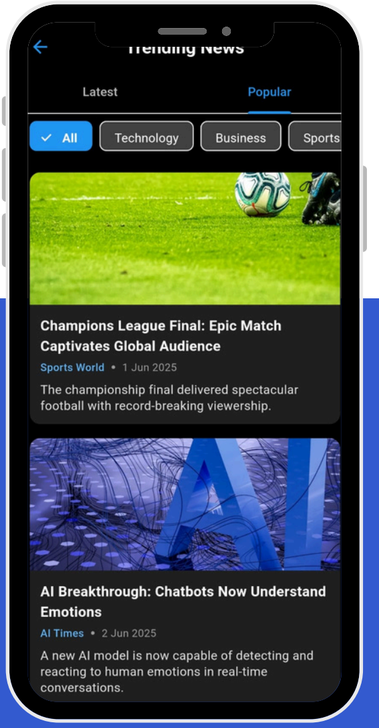

News App
Your personalized news assistant
Loading your news...


Welcome back!
Sign in to access your personalized news
Email
Password
Forgot password?
SIGN IN
Don't have an account? Sign Up

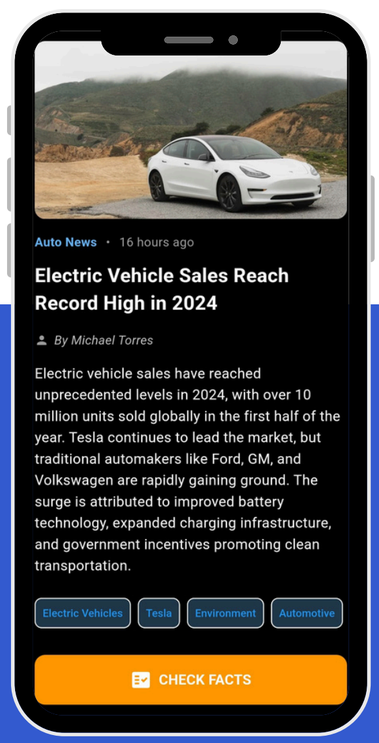

Create your account
Sign up to start your personalized news experience
Email
Password
SIGN UP
Already have an account? Sign In

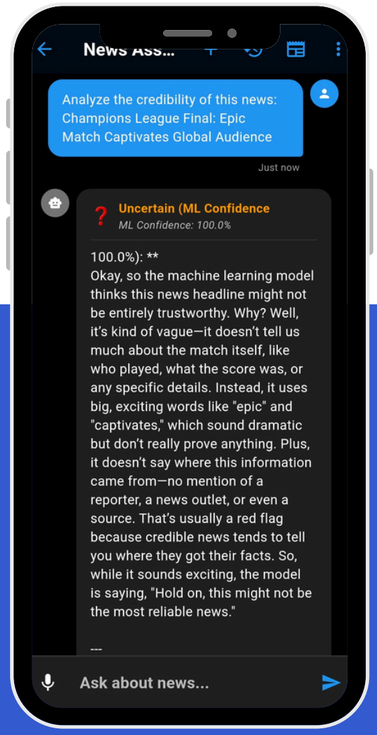

News Assistant
Hi! I'm your news assistant. I can help you verify if news stories are real or fake. Just share a news headline or claim with me!
12m ago
Ask about news...


Latest
Popular
All
Technology
Business
Sports
AI Breakthrough: Chatbots Now Understand Emotions
AI Times • 2 Jun 2025
A new AI model is now capable of detecting and reacting to human emotions in real-time conversations.
NASA Confirms Aliens Will Arrive by 2025
Alien Watch • 2 Jun 2025
A viral post claims NASA confirmed extraterrestrial beings are coming to Earth.

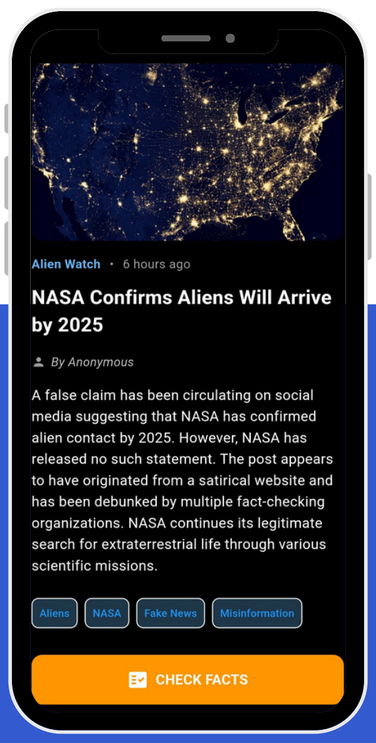

Latest
Popular
All
Technology
Business
Sports
Champions League Final: Epic Match Captivates Global Audience
Sports World • 1 Jun 2025
The championship final delivered spectacular football with record-breaking viewership.
AI Breakthrough: Chatbots Now Understand Emotions
AI Times • 2 Jun 2025
A new AI model is now capable of detecting and reacting to human emotions in real-time conversations.

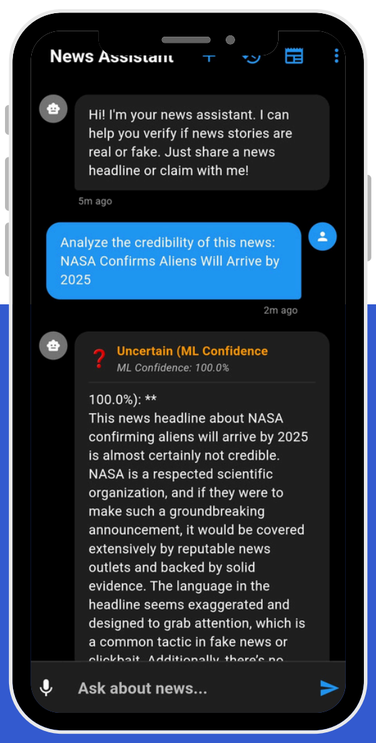
INTERFACE

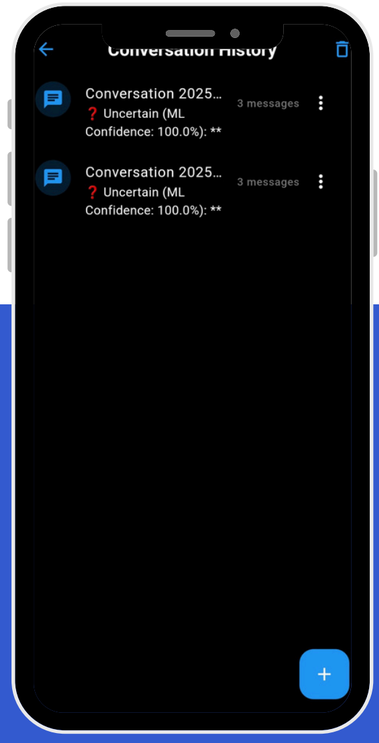

Auto News • 16 hours ago
Electric Vehicle Sales Reach Record High in 2024
By Michael Torres
Electric vehicle sales have reached unprecedented levels in 2024, with over 10 million units sold globally in the first half of the year. Tesla continues to lead the market, but traditional automakers like Ford, GM, and Volkswagen are rapidly gaining ground. The surge is attributed to improved battery technology, expanded charging infrastructure, and government incentives promoting clean transportation.
Electric Vehicles
Tesla
Environment
Automotive
CHECK FACTS

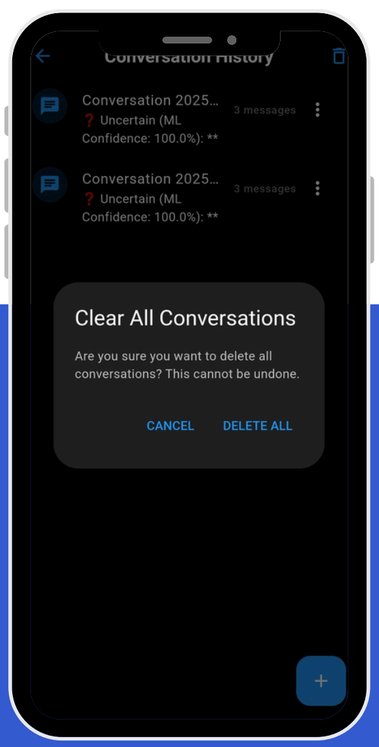

News Ass...
Analyze the credibility of this news: Champions League Final: Epic Match Captivates Global Audience
Just now
Uncertain (ML Confidence
ML Confidence: 100.0%
100.0%): **
Okay, so the machine learning model thinks this news headline might not be entirely trustworthy. Why? Well, it's kind of vague—it doesn't tell us much about the match itself, like who played, what the score was, or any specific details. Instead, it uses big, exciting words like "epic" and "captivates," which sound dramatic but don't really prove anything. Plus, it doesn't say where this information came from—no mention of a reporter, a news outlet, or even a source. That's usually a red flag because credible news tends to tell you where they got their facts. So, while it sounds exciting, the model is saying, "Hold on, this might not be the most reliable news."

Ask about news...

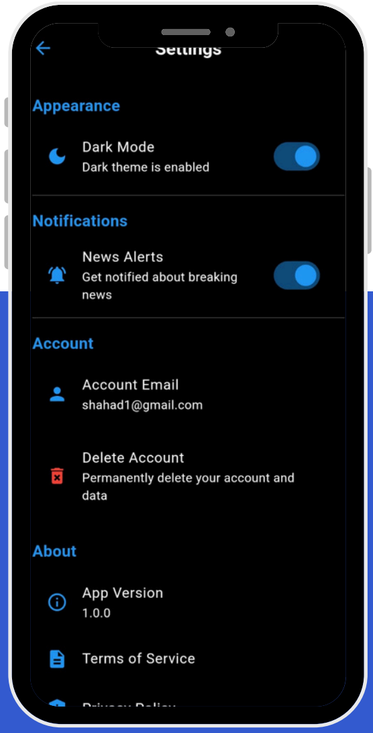

Alien Watch • 6 hours ago
NASA Confirms Aliens Will Arrive by 2025
By Anonymous
A false claim has been circulating on social media suggesting that NASA has confirmed alien contact by 2025. However, NASA has released no such statement. The post appears to have originated from a satirical website and has been debunked by multiple fact-checking organizations. NASA continues its legitimate search for extraterrestrial life through various scientific missions.
Aliens
NASA
Fake News
Misinformation
CHECK FACTS

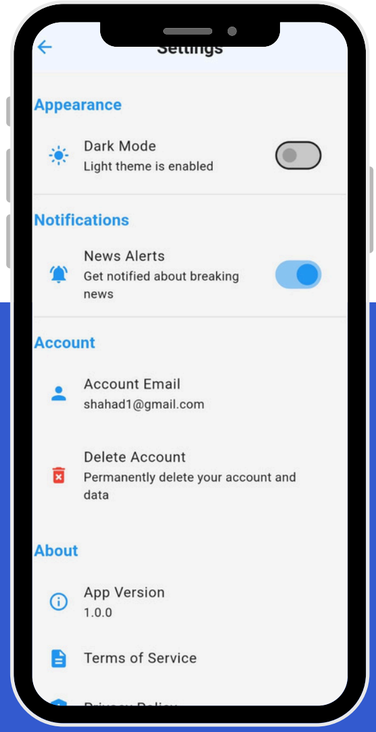

News Assistant
Hi! I'm your news assistant. I can help you verify if news stories are real or fake. Just share a news headline or claim with me!
5m ago
Analyze the credibility of this news: NASA Confirms Aliens Will Arrive by 2025
2m ago
Uncertain (ML Confidence
ML Confidence: 100.0%
100.0%): **
This news headline about NASA confirming aliens will arrive by 2025 is almost certainly not credible. NASA is a respected scientific organization, and if they were to make such a groundbreaking announcement, it would be covered extensively by reputable news outlets and backed by solid evidence. The language in the headline seems exaggerated and designed to grab attention, which is a common tactic in fake news or
Ask about news...


Conversation History
Conversation 2025...
Uncertain (ML Confidence: 100.0%): **
3 messages
Conversation 2025...
Uncertain (ML Confidence: 100.0%): **
3 messages

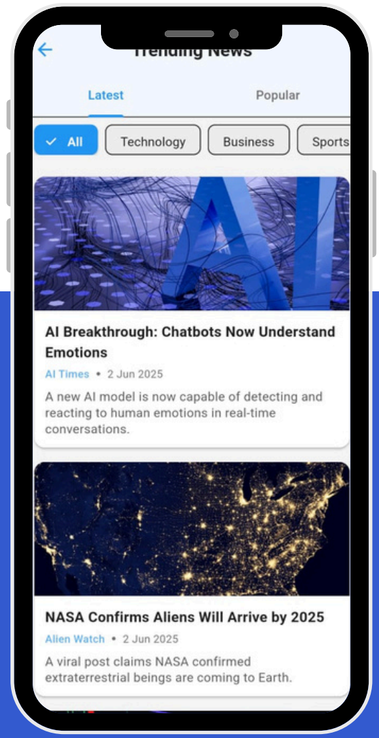

Conversation History
Conversation 2025...
Uncertain (ML Confidence: 100.0%): **
3 messages
Conversation 2025...
Uncertain (ML Confidence: 100.0%): **
3 messages
Clear All Conversations
Are you sure you want to delete all conversations? This cannot be undone.
CANCEL
DELETE ALL

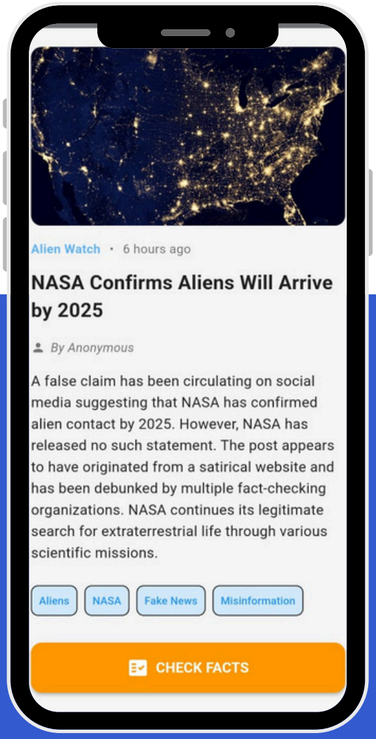
INTERFACE

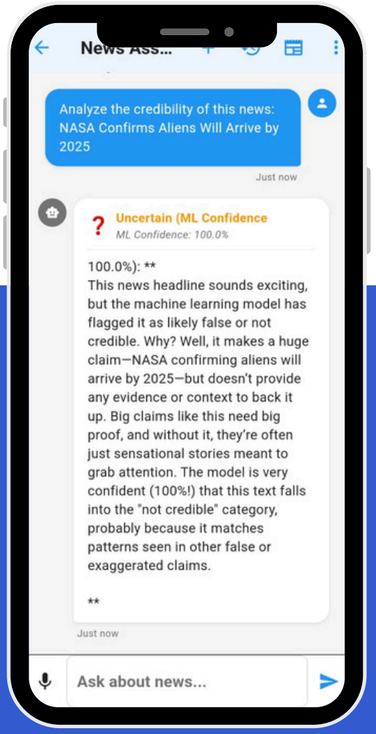

Settings
Appearance
Dark Mode
Dark theme is enabled
Notifications
News Alerts
Get notified about breaking news
Account
Account Email
shahad1@gmail.com
Delete Account
Permanently delete your account and data
About
App Version
1.0.0
Terms of Service

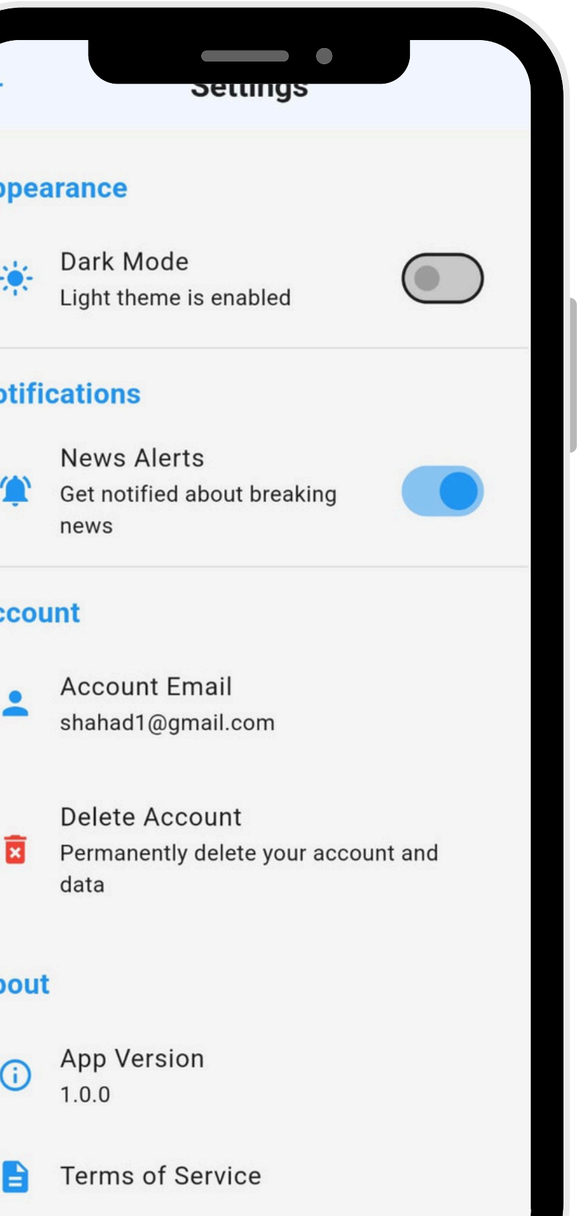

Settings
Appearance
Dark Mode
Light theme is enabled
Notifications
News Alerts
Get notified about breaking news
Account
Account Email
shahad1@gmail.com
Delete Account
Permanently delete your account and data
About
App Version
1.0.0
Terms of Service

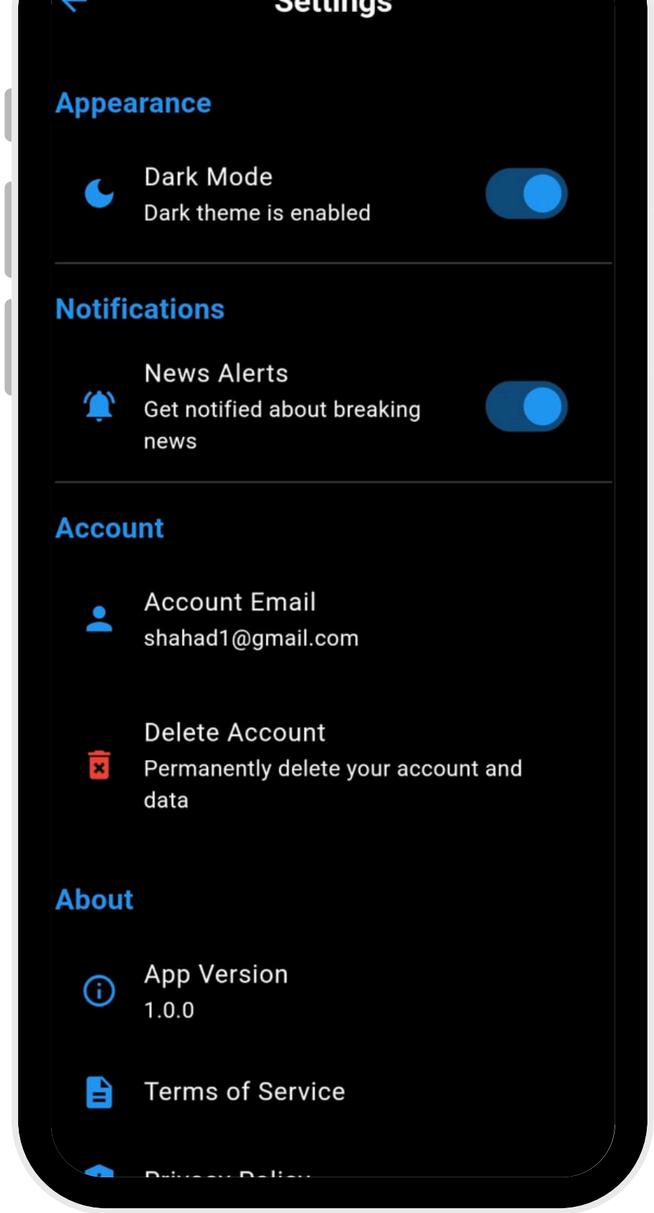

News Assistant
Hi! I'm your news assistant. I can help you verify if news stories are real or fake. Just share a news headline or claim with me!
4m ago
Ask about news...


Trending News
Latest
Popular
All
Technology
Business
Sports
AI Breakthrough: Chatbots Now Understand Emotions
AI Times • 2 Jun 2025
A new AI model is now capable of detecting and reacting to human emotions in real-time conversations.
NASA Confirms Aliens Will Arrive by 2025
Alien Watch • 2 Jun 2025
A viral post claims NASA confirmed extraterrestrial beings are coming to Earth.


Alien Watch • 6 hours ago
NASA Confirms Aliens Will Arrive by 2025
By Anonymous
A false claim has been circulating on social media suggesting that NASA has confirmed alien contact by 2025. However, NASA has released no such statement. The post appears to have originated from a satirical website and has been debunked by multiple fact-checking organizations. NASA continues its legitimate search for extraterrestrial life through various scientific missions.
Aliens
NASA
Fake News
Misinformation
CHECK FACTS


News Ass...
Analyze the credibility of this news: NASA Confirms Aliens Will Arrive by 2025
Just now
Uncertain (ML Confidence
ML Confidence: 100.0%
100.0%): **
This news headline sounds exciting, but the machine learning model has flagged it as likely false or not credible. Why? Well, it makes a huge claim—NASA confirming aliens will arrive by 2025—but doesn't provide any evidence or context to back it up. Big claims like this need big proof, and without it, they're often just sensational stories meant to grab attention. The model is very confident (100%!) that this text falls into the "not credible" category, probably because it matches patterns seen in other false or exaggerated claims.
**
Just now
Ask about news...





Features of the App News Assistant (main screen)

Trending section

Conversation History

Smart credibility analysis

Dark and light mode

Friendly UI/UX

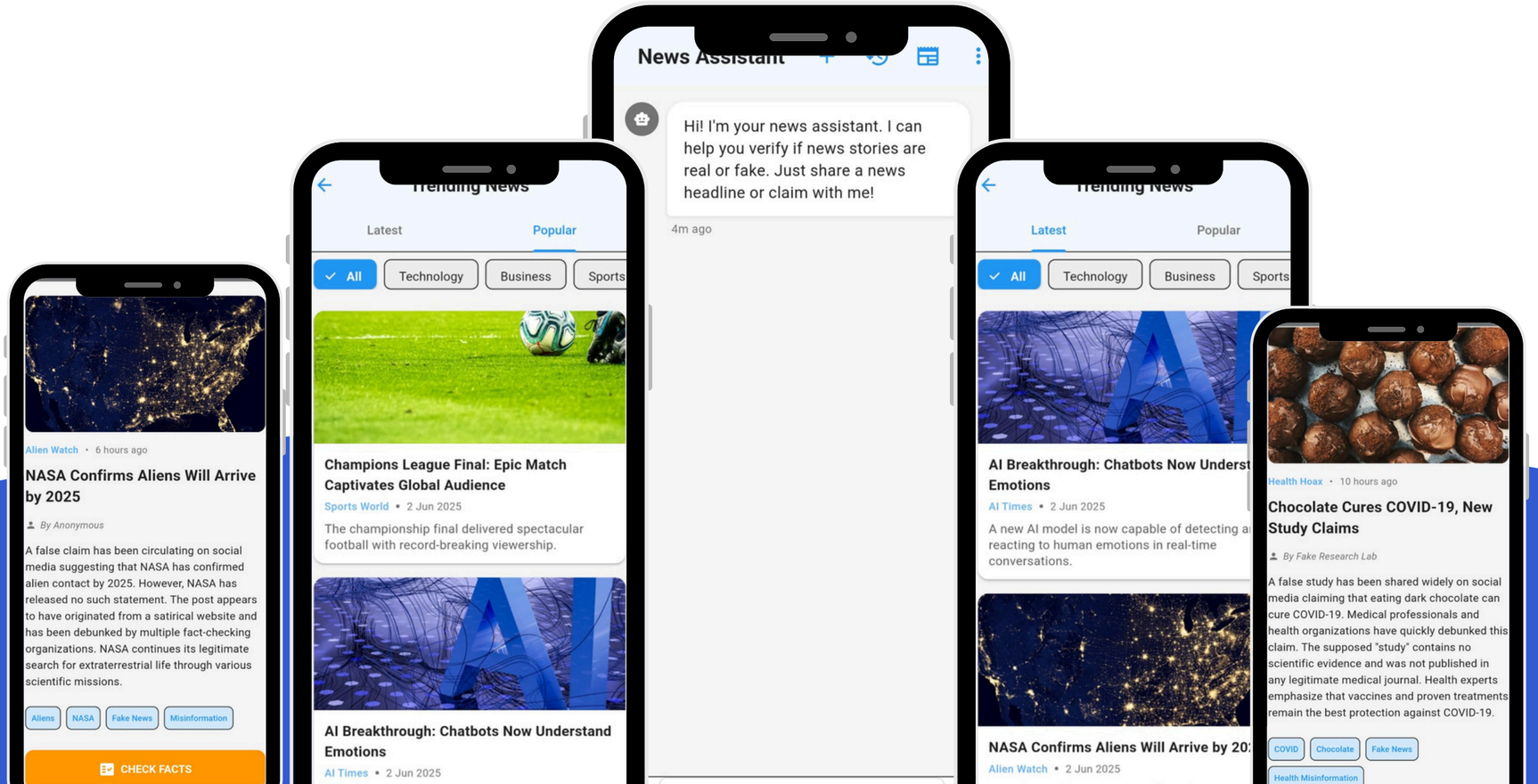
Profile section (loads news instead of logging out)

Testing

Feature	Test Status	Result
User Login	Passed	Working correctly
User Signup	Passed	Working correctly
News Submission	Passed	Input accepted
Fake News Classification via API	Passed	Results received
Conversation History Saved in Firebase	Passed	Data stored properly
Settings Page	Passed	Modifiable settings

Table 2 Testing

Thank You


News Assistant
Hi! I'm your news assistant. I can help you verify if news stories are real or fake. Just share a news headline or claim with me!
4m ago
Trending News
Latest
Popular
All
Technology
Business
Sports
Champions League Final: Epic Match Captivates Global Audience
Sports World • 2 Jun 2025
The championship final delivered spectacular football with record-breaking viewership.
AI Breakthrough: Chatbots Now Understand Emotions
AI Times • 2 Jun 2025
Alien Watch • 6 hours ago
NASA Confirms Aliens Will Arrive by 2025
By Anonymous
A false claim has been circulating on social media suggesting that NASA has confirmed alien contact by 2025. However, NASA has released no such statement. The post appears to have originated from a satirical website and has been debunked by multiple fact-checking organizations. NASA continues its legitimate search for extraterrestrial life through various scientific missions.
Aliens
NASA
Fake News
Misinformation
CHECK FACTS
Health Hoax • 10 hours ago
Chocolate Cures COVID-19, New Study Claims
By Fake Research Lab
A false study has been shared widely on social media claiming that eating dark chocolate can cure COVID-19. Medical professionals and health organizations have quickly debunked this claim. The supposed "study" contains no scientific evidence and was not published in any legitimate medical journal. Health experts emphasize that vaccines and proven treatments remain the best protection against COVID-19.
COVID
Chocolate
Fake News
Health Misinformation